Trust Points and AHP Guide

Agrivive flagging. Simple words. Sample numbers made by script, so math is right.

1. Why this paper

Points now are own picks. No study give them. Panel say: need a source. Fix: get points from **AHP**. AHP is old, known way to turn judge opinion into numbers. Judges compare two things at a time. Math make weights. Math also check judges not mixed up.

2. What AHP is, short

AHP = Analytic Hierarchy Process. Made by Thomas Saaty (1980). Big idea: people bad at ranking 15 things at once. People good at comparing 2 things. So ask only pairs. Let math do the rest.

Saaty scale	Meaning
1	Same
3	A bit worse
5	Clearly worse
7	Much worse
9	Extreme
2, 4, 6, 8	In between
1/3, 1/5 ...	Reverse. Second thing is worse.

3. List of violations and points now

Points below are **current own picks** from `backend/src/utils/trust/policy.ts`. AHP below will replace them with points that have a source.

Violation	Who	Points now	What else
Inquiry non-response	Seller	1	Reminder at 12h, warning at 24h
Low rating (2 stars or less)	Seller	1	Counts with a reason or 2nd in 30 days
Seller cancels confirmed order	Seller	2	Also adds to fault rate
Misleading product info	Seller	2	Also adds to fault rate
Inappropriate communication	Seller	2	Needs 2 different reporters
Seller-fault decline at pickup	Seller	2	Also adds to fault rate
QR misuse	Seller	3	**Block at once**
Missed pickup	Buyer	2	Always 2 points
Late cancel	Buyer	1	Cancel counter warns at 5
Invalid decline at pickup	Buyer	1	Dispute decided by system
False decline photo	Buyer	2	
Fake reports	Buyer	1	
Order from own account	Buyer	3	**Block at once**
Chat spam	Both	1	
Personal data in chat	Both	1	
Review bomb	Buyer	none	Own rule, no points
Order flood	Buyer	none	Own rule, no points

4. Trust levels and consequences (same for buyer and seller)

Score	Level	Consequence
0-1	Good standing	Nothing
2-3	Notice	Notice sent
4-5	Warning	Warning sent
6-7	Pause 24 hours	Account paused 24 hours
8-11	Pause 3 days	Account paused 72 hours
12+	Blocked	Account closed (isActive false)

A penalty starts only when a person moves UP a level. Check again later, no second penalty. Serious violation skip steps and block at once. Seller pause or block also needs fault rate 10% or more over 10 or more orders. Points fade after 90 days.

5. Formulas

Name	Formula
Trust score (Eq. 1)	S = sum of weights of verified events in last 90 days. Excused or overturned events skip.
Seller fault rate (Eq. 2)	F = (seller cancels + confirmed wrong listings + upheld declines) / (completed + cancelled + declined orders) x 100
AHP weight	Divide each matrix cell by its column sum. Weight = average of each row.
Largest eigenvalue	lambda max = average of ((matrix x weights) / weights)
Consistency index	CI = (lambda max - n) / (n - 1)
Consistency ratio	CR = CI / RI. Keep if CR is 0.10 or less. RI: n=3 0.58, n=4 0.90, n=5 1.12 (Saaty).
Points	Points = round(weight / smallest weight), at least 1
Many judges	Merged cell = geometric mean of each judge cell

6. How to use AHP in Agrivive

Step	Do
1	Split the 15 violations into groups: pickup, listing, chat, review. One big matrix is too hard.
2	Pick 3 to 5 judges: adviser, City Agriculturist's office, some sellers, some buyers.
3	Each judge fills a pair matrix per group with the Saaty scale.
4	Merge judges with the geometric mean.
5	Run AHP. Get weights and CR. If CR is over 0.10, judges redo the pairs.
6	Run group-level AHP. Final weight = group weight x weight inside group.
7	Turn weights into points. Set thresholds as fractions of total possible score.
8	Put numbers in `trustPolicyWeights` in policy.ts. Put matrices and CR in the paper.

7. Five samples with process

Each sample shows: judge pairs, matrix, column sums, weights, consistency check, points. All numbers made by script. Judge answers are **sample data**.

Sample 1. Seller violations

Judge ask: which fault is worse for the market? QR misuse is a bit worse than a seller cancel (3), clearly worse than a slow reply (5), much worse than chat spam (7).

Sample data. Not real judge answers.

Step 1. Pair matrix

	QR misuse	Seller cancels order	Inquiry non-response	Chat spam
QR misuse	1	3	5	7
Seller cancels order	1/3	1	3	5
Inquiry non-response	1/5	1/3	1	3
Chat spam	1/7	1/5	1/3	1
Column sum	1.676	4.533	9.333	16.000

Step 2. Divide each cell by its column sum. Average each row. Row average = weight.

	QR misuse	Seller cancels order	Inquiry non-response	Chat spam	Weight
QR misuse	0.597	0.662	0.536	0.438	**0.558**
Seller cancels order	0.199	0.221	0.321	0.312	**0.263**
Inquiry non-response	0.119	0.074	0.107	0.188	**0.122**
Chat spam	0.085	0.044	0.036	0.062	**0.057**

Step 3. Check judges did not fight themselves

lambda max = 4.118. CI = (lambda max - n) / (n - 1) = 0.0395. CR = CI / RI = 0.0395 / 0.90 = **0.044**. Under 0.10. Keep.

Step 4. Make points

Violation	Weight	Weight / smallest weight	Points
QR misuse	0.558	9.81	10
Seller cancels order	0.263	4.63	5
Inquiry non-response	0.122	2.14	2
Chat spam	0.057	1.00	1

Sample 2. Buyer violations

Missed pickup hurts the seller most: stall food waits and nobody comes. A late cancel hurts least.

Sample data. Not real judge answers.

Step 1. Pair matrix

	Missed pickup	False decline photo	Invalid decline	Late cancel
Missed pickup	1	2	3	5
False decline photo	1/2	1	2	3
Invalid decline	1/3	1/2	1	2
Late cancel	1/5	1/3	1/2	1
Column sum	2.033	3.833	6.500	11.000

Step 2. Divide each cell by its column sum. Average each row. Row average = weight.

	Missed pickup	False decline photo	Invalid decline	Late cancel	Weight
Missed pickup	0.492	0.522	0.462	0.455	**0.482**
False decline photo	0.246	0.261	0.308	0.273	**0.272**
Invalid decline	0.164	0.130	0.154	0.182	**0.158**
Late cancel	0.098	0.087	0.077	0.091	**0.088**

Step 3. Check judges did not fight themselves

lambda max = 4.015. CI = (lambda max - n) / (n - 1) = 0.0048. CR = CI / RI = 0.0048 / 0.90 = **0.005**. Under 0.10. Keep.

Step 4. Make points

Violation	Weight	Weight / smallest weight	Points
Missed pickup	0.482	5.46	5
False decline photo	0.272	3.08	3
Invalid decline	0.158	1.78	2
Late cancel	0.088	1.00	1

Sample 3. Chat violations (3 items)

Small group, 3 pairs only. Easy for a judge. Insults are worse than sharing a phone number, which is worse than spam.

Sample data. Not real judge answers.

Step 1. Pair matrix

	Inappropriate talk	Personal data in chat	Chat spam
Inappropriate talk	1	3	5
Personal data in chat	1/3	1	3
Chat spam	1/5	1/3	1
Column sum	1.533	4.333	9.000

Step 2. Divide each cell by its column sum. Average each row. Row average = weight.

	Inappropriate talk	Personal data in chat	Chat spam	Weight
Inappropriate talk	0.652	0.692	0.556	**0.633**
Personal data in chat	0.217	0.231	0.333	**0.260**
Chat spam	0.130	0.077	0.111	**0.106**

Step 3. Check judges did not fight themselves

lambda max = 3.039. CI = (lambda max - n) / (n - 1) = 0.0194. CR = CI / RI = 0.0194 / 0.58 = **0.033**. Under 0.10. Keep.

Step 4. Make points

Violation	Weight	Weight / smallest weight	Points
Inappropriate talk	0.633	5.97	6
Personal data in chat	0.260	2.45	2
Chat spam	0.106	1.00	1

Sample 4. Group level (which group matters more)

One 15 x 15 matrix is too big. So run AHP inside each group, then run this small AHP on top. Final weight of a violation = group weight x its weight inside the group.

Sample data. Not real judge answers.

Step 1. Pair matrix

	Pickup group	Listing group	Chat group	Review group
Pickup group	1	2	4	3
Listing group	1/2	1	3	2
Chat group	1/4	1/3	1	1/2
Review group	1/3	1/2	2	1
Column sum	2.083	3.833	10.000	6.500

Step 2. Divide each cell by its column sum. Average each row. Row average = weight.

	Pickup group	Listing group	Chat group	Review group	Weight
Pickup group	0.480	0.522	0.400	0.462	**0.466**
Listing group	0.240	0.261	0.300	0.308	**0.277**
Chat group	0.120	0.087	0.100	0.077	**0.096**
Review group	0.160	0.130	0.200	0.154	**0.161**

Step 3. Check judges did not fight themselves

lambda max = 4.031. CI = (lambda max - n) / (n - 1) = 0.0103. CR = CI / RI = 0.0103 / 0.90 = **0.011**. Under 0.10. Keep.

Step 4. Make points

Violation	Weight	Weight / smallest weight	Points
Pickup group	0.466	4.85	5
Listing group	0.277	2.89	3
Chat group	0.096	1.00	1
Review group	0.161	1.68	2

Sample 5. Two judges combined

Judge 1 and Judge 2 each fill a matrix. Merge each cell with the geometric mean. Then run AHP once on the merged matrix.

Sample data. Not real judge answers.

Step 0. Judge 1 matrix

	QR misuse	Seller cancels order	Chat spam
QR misuse	1	3	7
Seller cancels order	1/3	1	3
Chat spam	1/7	1/3	1

Step 0. Judge 2 matrix

	QR misuse	Seller cancels order	Chat spam
QR misuse	1	5	9
Seller cancels order	1/5	1	3
Chat spam	1/9	1/3	1

Step 1. Pair matrix (merged, geometric mean)

	QR misuse	Seller cancels order	Chat spam
QR misuse	1	3.873	7.937
Seller cancels order	0.258	1	3
Chat spam	0.126	1/3	1
Column sum	1.384	5.206	11.937

Step 2. Divide each cell by its column sum. Average each row. Row average = weight.

	QR misuse	Seller cancels order	Chat spam	Weight
QR misuse	0.722	0.744	0.665	**0.710**
Seller cancels order	0.187	0.192	0.251	**0.210**
Chat spam	0.091	0.064	0.084	**0.080**

Step 3. Check judges did not fight themselves

lambda max = 3.016. CI = (lambda max - n) / (n - 1) = 0.0081. CR = CI / RI = 0.0081 / 0.58 = **0.014**. Under 0.10. Keep.

Step 4. Make points

Violation	Weight	Weight / smallest weight	Points
QR misuse	0.710	8.92	9

Violation	Weight	Weight / smallest weight	Points
Seller cancels order	0.210	2.64	3
Chat spam	0.080	1.00	1

Honest limits

AHP give a **method** with a source. It does not give judge answers. Reference is the method plus the record of who judged and what CR came out. Write it that way in the paper. Do not say a study gave '3 points for QR misuse'.

Sample numbers here are NOT real. Real judges must fill the matrices.

Sources to read in full before citing: Saaty, T. L. (1980), The Analytic Hierarchy Process, McGraw-Hill. Dodevska et al., Mathematics 2023, 11(3):604. Systems 2025, 13(5):328. Structure of weighted trust points: Kokkodis, Lappas and Kane (2022); Chu and Chen (2022); OECD (2022); WIPO ACE/16/11 (2023).

Final points and thresholds still need beta testing, then tune.